

Ex.No: 5	DATA VISUALIZATION USING MATPLOTLIB
Date:	

AIM:

ALGORITHM:

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PROGRAM:

```
import matplotlib.pyplot as plt

months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]
sales = [120, 150, 180, 170, 220, 250]

plt.plot(months, sales, marker='o')
plt.title("Monthly Sales")
plt.xlabel("Months")
plt.ylabel("Sales")
plt.show()
```

```
plt.bar(months, sales, color='orange')
plt.title("Monthly Sales")
plt.xlabel("Months")
plt.ylabel("Sales")
plt.show()
```

```
departments = ["HR", "IT", "Finance", "Sales"]
employees = [25, 40, 20, 15]

plt.pie(employees, labels=departments, autopct='%1.1f%%', startangle=90)
plt.title("Employee Distribution")
plt.show()
```

```
marks = [45, 55, 60, 70, 75, 80, 85, 90, 95, 100, 65, 70, 75, 80, 85]

plt.hist(marks, bins=5, color='skyblue', edgecolor='black')
plt.title("Student Marks Distribution")
plt.xlabel("Marks Range")
plt.ylabel("Frequency")
plt.show()
```

```
hours = [2, 3, 4, 5, 6, 7, 8, 9, 10]
marks = [50, 55, 60, 65, 70, 75, 80, 85, 90]

plt.scatter(hours, marks, color='red')
plt.title("Study Hours vs Marks")
plt.xlabel("Study Hours")
plt.ylabel("Marks")
plt.show()
```

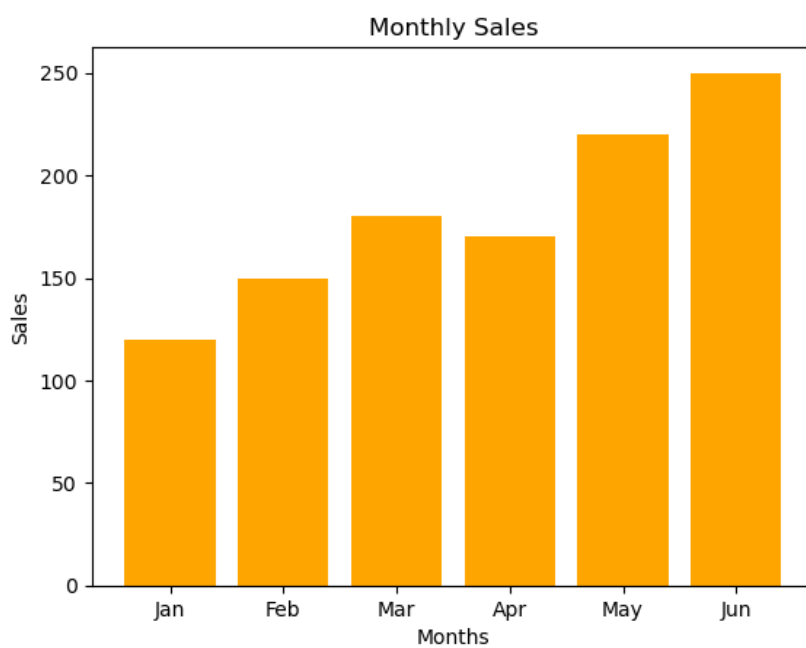
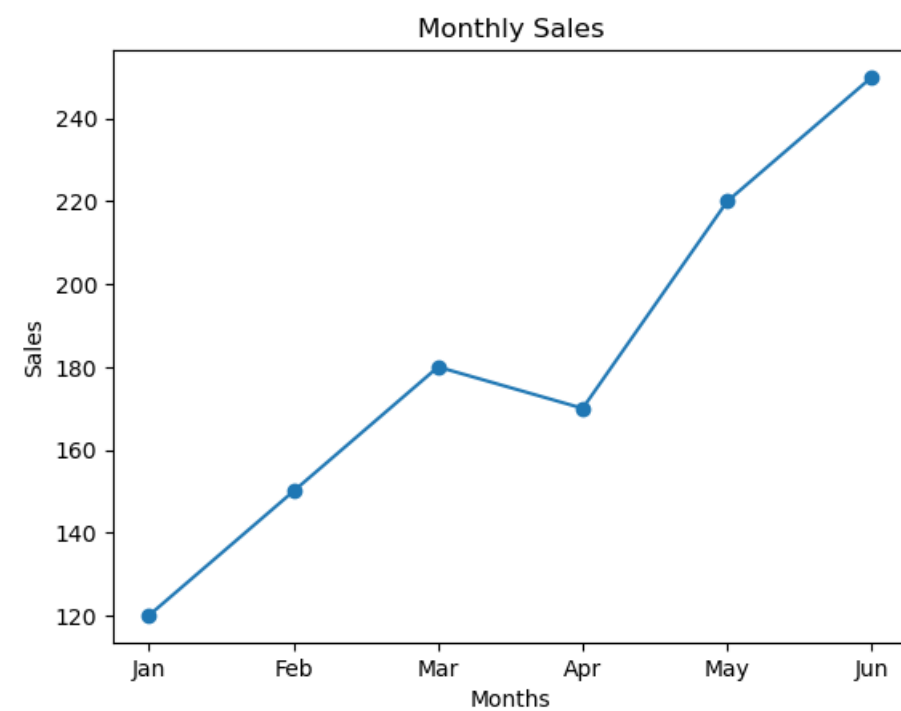
```
salary = [25000, 27000, 30000, 32000, 35000, 40000, 150000]

plt.boxplot(salary)
plt.title("Salary Distribution")
plt.show()
```

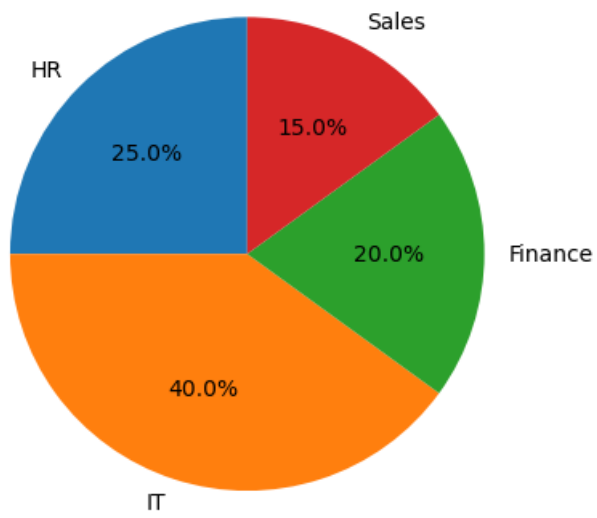
```
months = ["Jan","Feb","Mar","Apr"]
sales = [120,150,180,200]
profit = [20,30,40,50]

plt.plot(months, sales, label="Sales", marker='o')
plt.plot(months, profit, label="Profit", marker='s')
plt.title("Sales vs Profit Trend")
plt.xlabel("Months")
plt.ylabel("Values")
plt.legend()
plt.show()
```

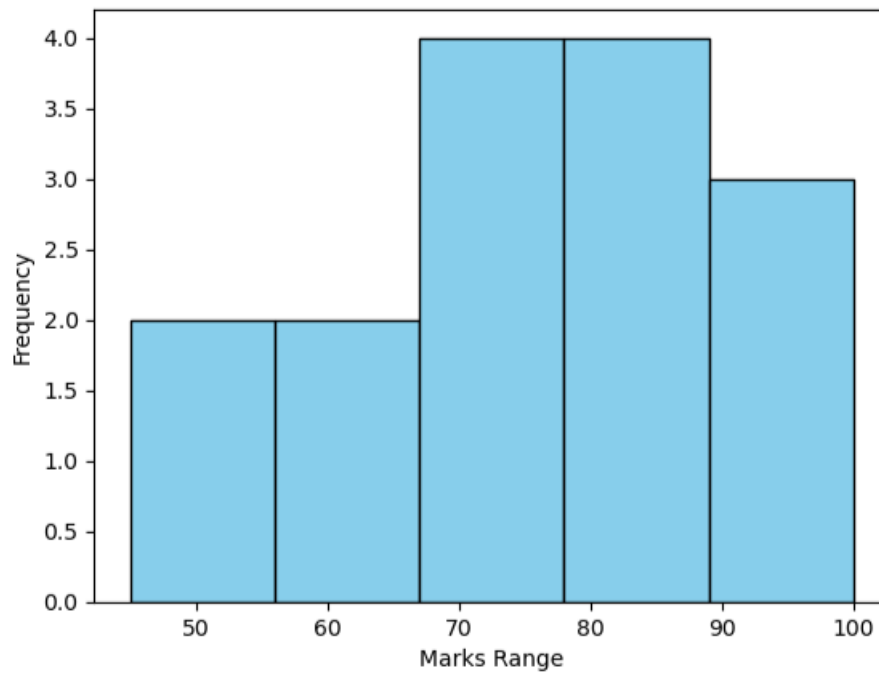
OUTPUT:

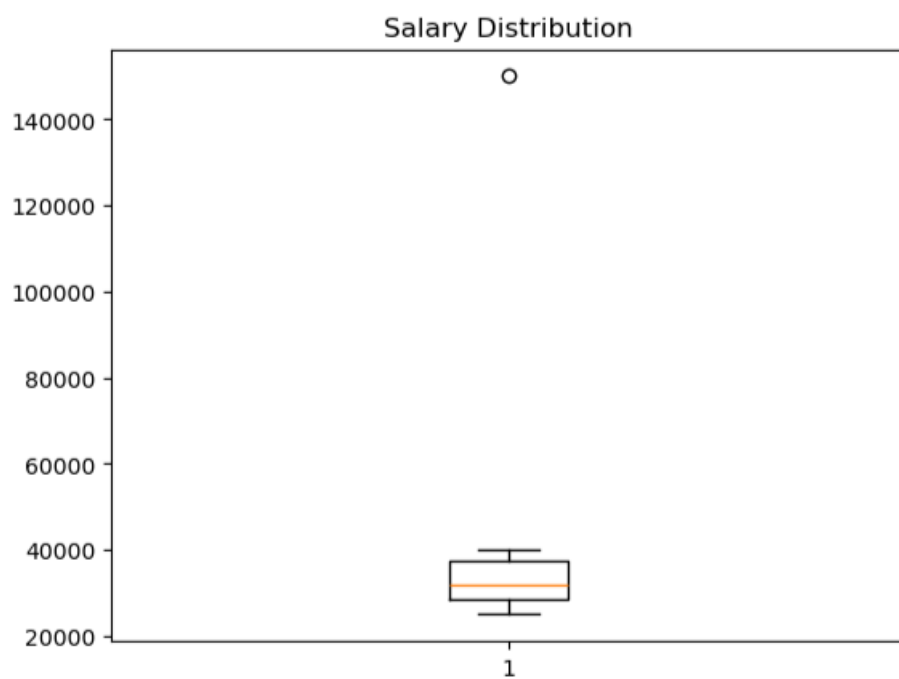
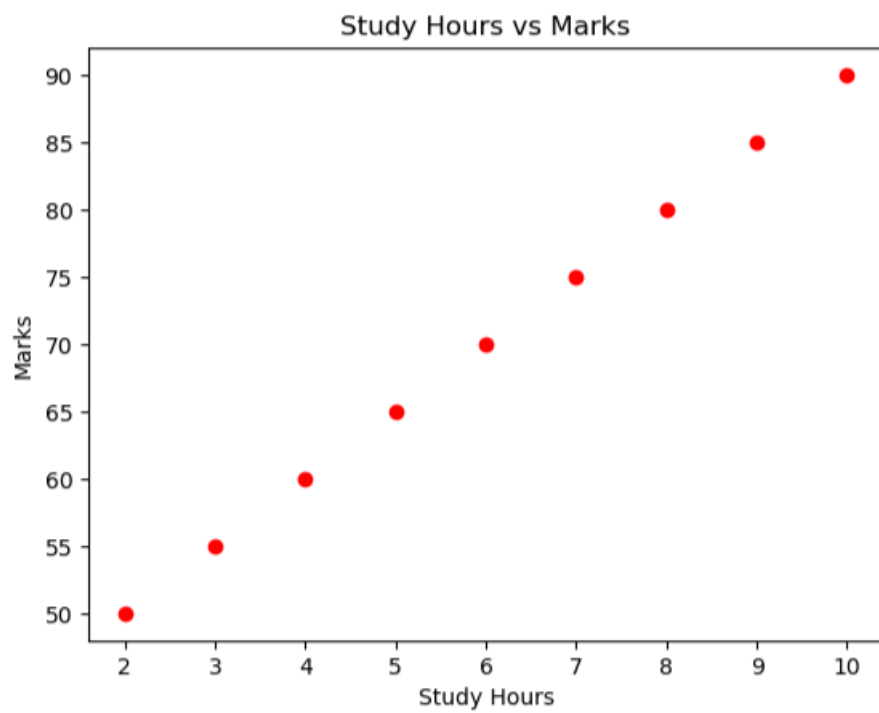


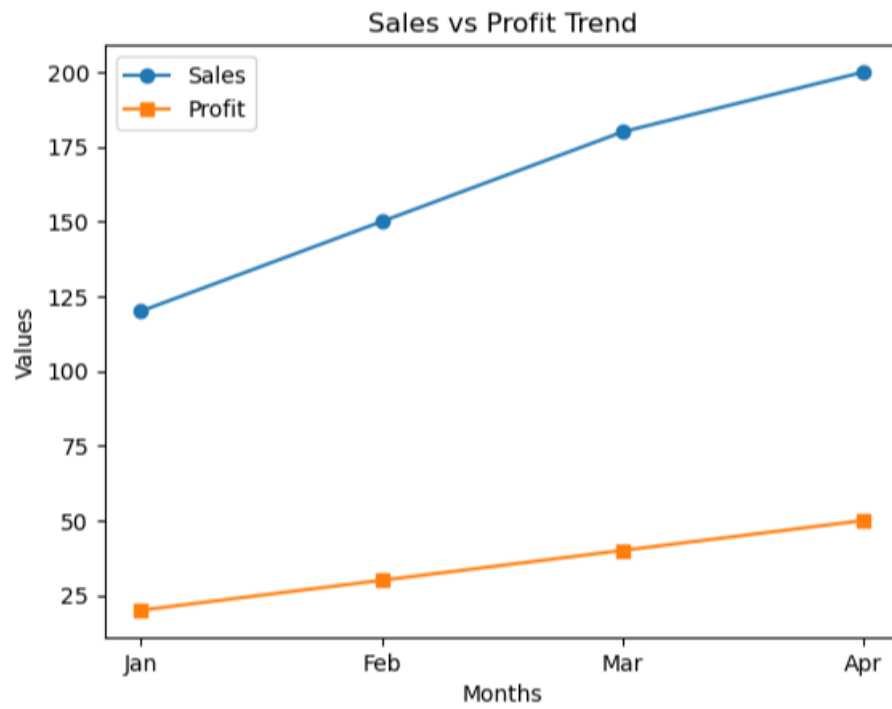
Employee Distribution



Student Marks Distribution







RESULT: